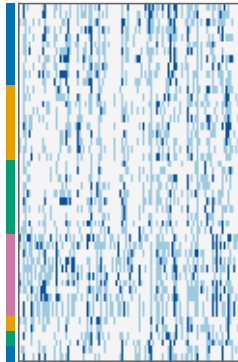


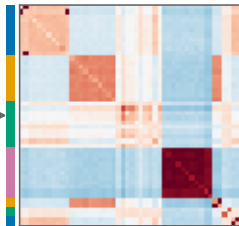
frkin: fixed rank + kinship threshold on the CS matrix

genotypes G



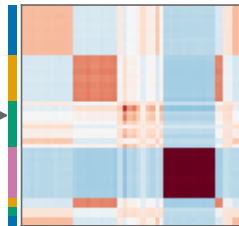
one pass

CS matrix H



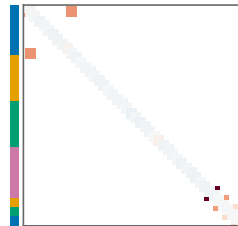
fixed rank
+ kinship

L : rank $k + 1$, diag. free



+

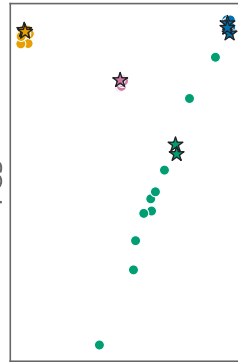
S : diagonal
+ related pairs



eigen-
vectors

PC3

PCs of L



PC2

● CEU ● CHB ● MXL ● YRI ★ relative